

Practice - Three-Product Production Plan

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:54

A workshop makes products A, B, C (X_1 , X_2 , X_3 units/week). Profit per unit 30, 40, 35. Machining, assembly and material capacities are limited, and a contract requires at least 10 units of B.

Model

Maximize $Z = 30X_1 + 40X_2 + 35X_3$
subject to
[Machining (h)] $2X_1 + 3X_2 + 2X_3 \leq 120$
[Assembly (h)] $4X_1 + 2X_2 + 3X_3 \leq 160$
[Material (kg)] $3X_1 + 2X_2 + 4X_3 \leq 140$
[Contract B] $X_2 \geq 10$
 $X_1, X_2, X_3 \geq 0$

RESULT

OPTIMAL

$Z^* = 1787.5$

$X_1 = 0$

$X_2 = 25$

$X_3 = 22.5$

iterations: 3 · rule: dantzig

Findings

- Optimal: row 0 has no negative coefficients. $Z^* = 1787.5$

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. Machining (h) (Type I, $\leq$): add slack $X_4 \geq 0$ (unused resource, cost 0) $\rightarrow$ X_4 is the starting basic variable of this row.
2. Assembly (h) (Type I, $\leq$): add slack $X_5 \geq 0$ (unused resource, cost 0) $\rightarrow$ X_5 is the starting basic variable of this row.
3. Material (kg) (Type I, $\leq$): add slack $X_6 \geq 0$ (unused resource, cost 0) $\rightarrow$ X_6 is the starting basic variable of this row.
4. Contract B (Type II, $\geq$): subtract surplus $X_7 \geq 0$ (cost 0). Its column is -1 , so it cannot start in the basis $\rightarrow$ an artificial variable is needed.
5. Contract B: add artificial $X_8 \geq 0$ with objective coefficient $-M$ (huge penalty) $\rightarrow$ it provides an immediately recognisable starting basic variable.

Maximize $Z = 30X_1 + 40X_2 + 35X_3 - MX_8$

subject to

$$2X_1 + 3X_2 + 2X_3 + X_4 = 120$$

$$4X_1 + 2X_2 + 3X_3 + X_5 = 160$$

$$3X_1 + 2X_2 + 4X_3 + X_6 = 140$$

$$X_2 - X_7 + X_8 = 10$$

$$X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8 \geq 0$$

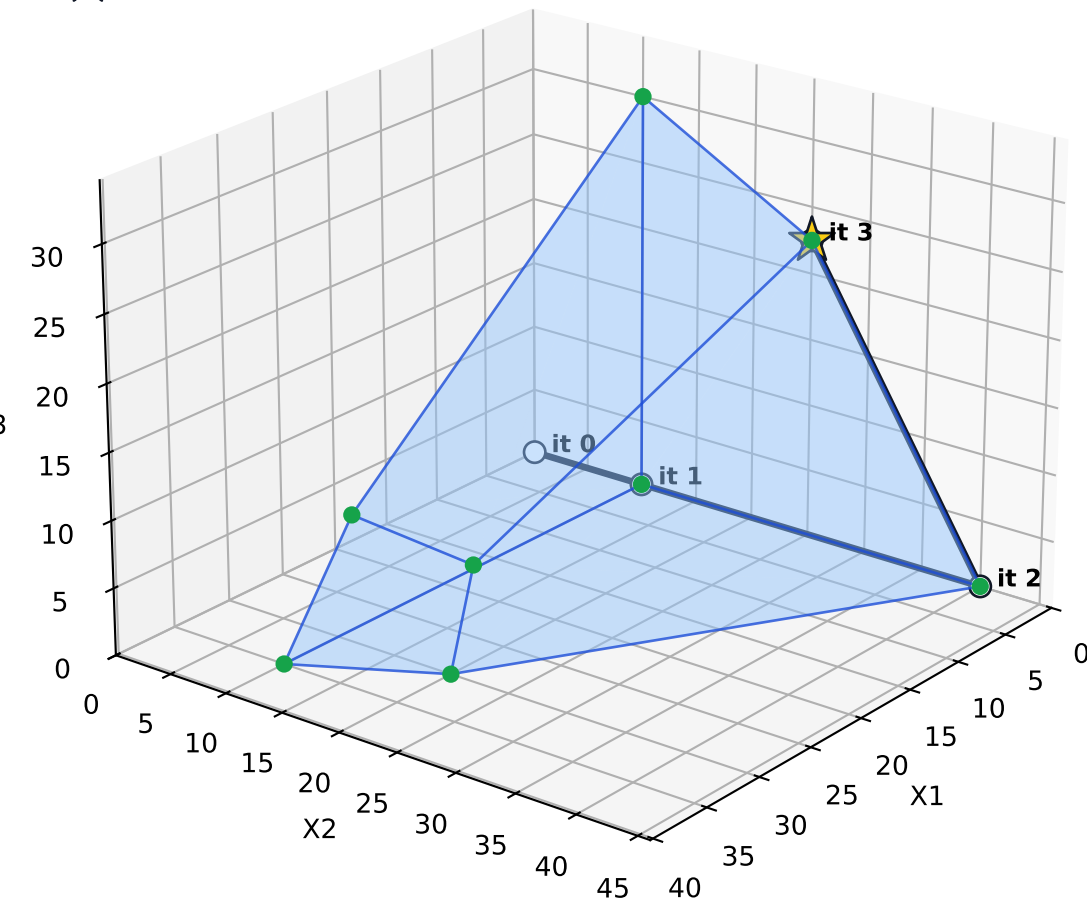
Standard form - coefficient matrix

	X1	X2	X3	X4	X5	X6	X7	X8		RHS
max Z	30	40	35	0	0	0	0	-M		
Machining (h)	2	3	2	1	0	0	0	0	=	120
Assembly (h)	4	2	3	0	1	0	0	0	=	160
Material (kg)	3	2	4	0	0	1	0	0	=	140
Contract B	0	1	0	0	0	0	-1	1	=	10

2. Geometric view: basic solutions, extreme points and the simplex path

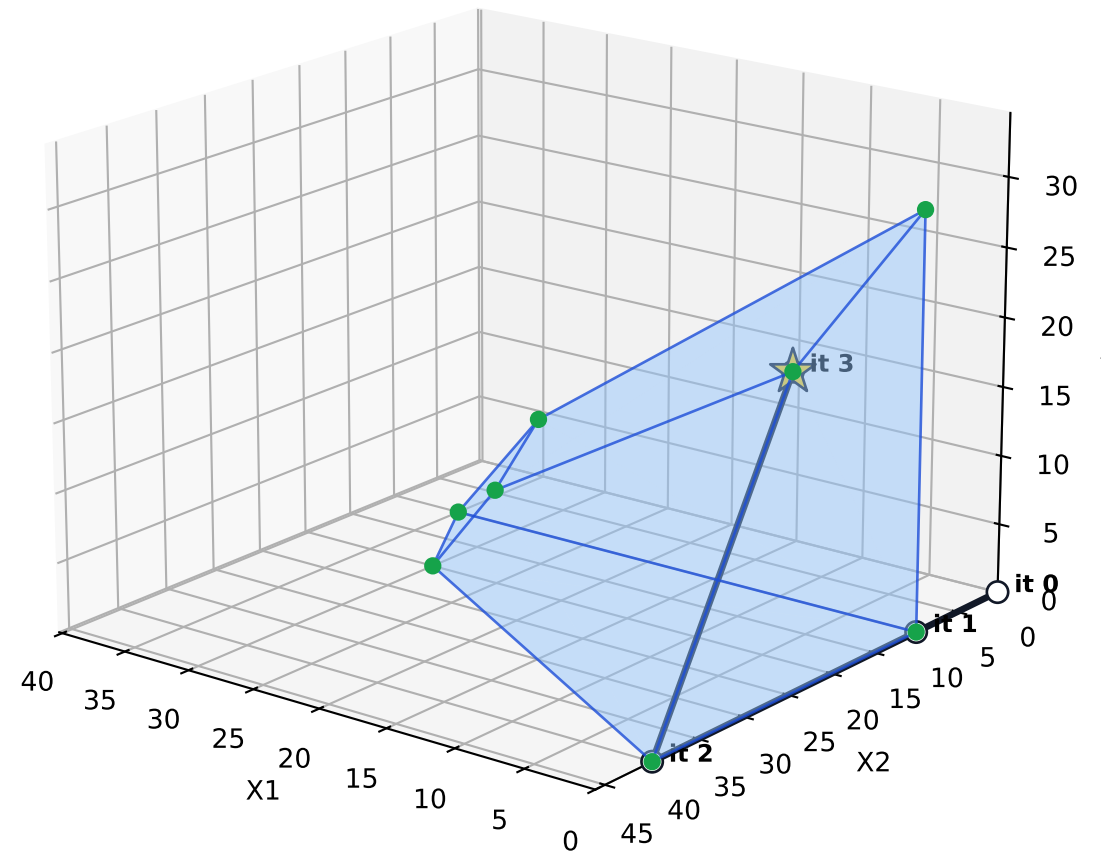
Feasible polytope & simplex path - optimal

- extreme points
- simplex path
- ★ optimum



...same polytope, rotated view

- extreme points
- simplex path
- ★ optimum



Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS
Z	0	1	-30	-40	-35	0	0	0	0	M	0
X4	1	0	2	3	2	1	0	0	0	0	120
X5	2	0	4	2	3	0	1	0	0	0	160
X6	3	0	3	2	4	0	0	1	0	0	140
X8	4	0	0	1	0	0	0	0	-1	1	10

Initial tableau. Basic variables: $X4 = 120$, $X5 = 160$, $X6 = 140$, $X8 = 10$. Non-basic (set to 0): $X1$, $X2$, $X3$, $X7$. The values of the basic variables are read directly from the RHS column.

Initial tableau after Big-M preparation

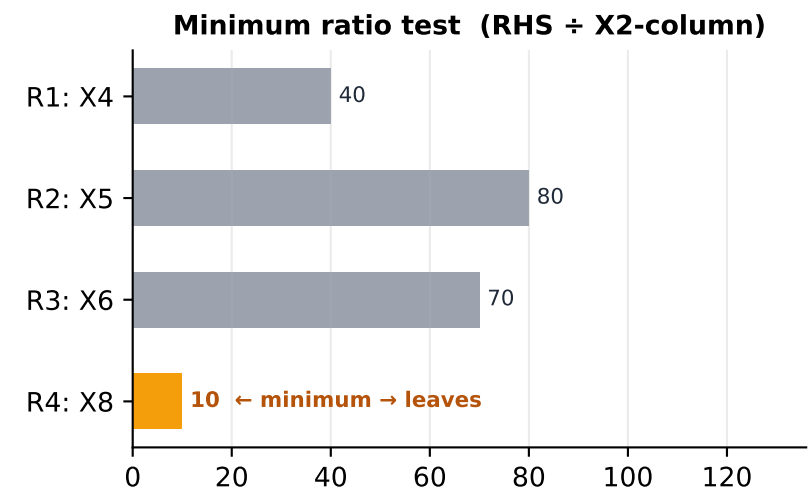
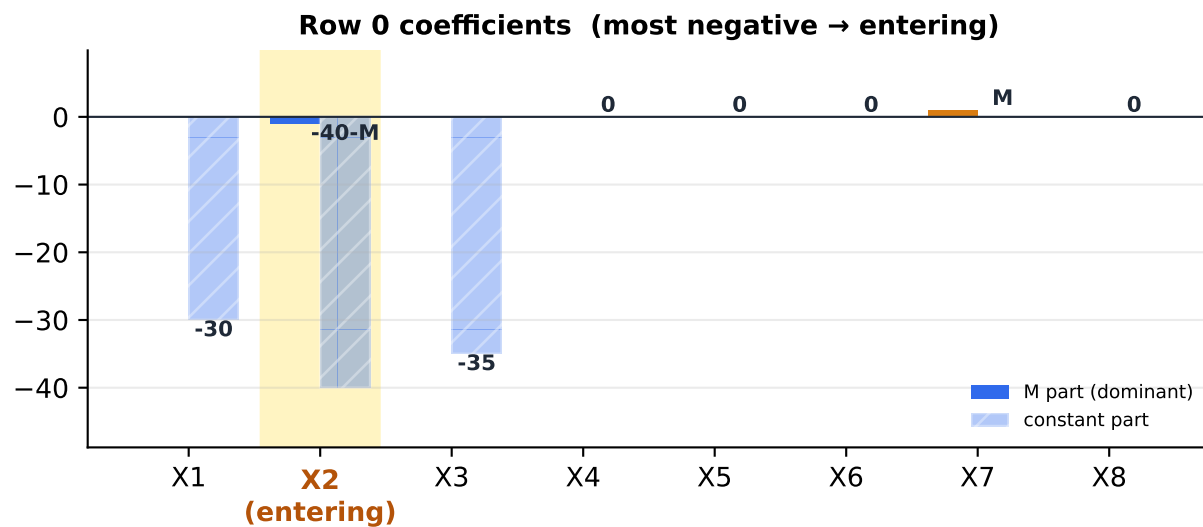
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS
Z	0	1	-30	-40-M	-35	0	0	0	M	0	-10M
X4	1	0	2	3	2	1	0	0	0	0	120
X5	2	0	4	2	3	0	1	0	0	0	160
X6	3	0	3	2	4	0	0	1	0	0	140
X8	4	0	0	1	0	0	0	0	-1	1	10

Big-M preparation. the artificial basic variable(s) had coefficient M in row 0. Subtracting M × their rows makes those coefficients zero, so row 0 is ready for the optimality test.

- Subtract M × R4 from R0 (make the coefficient of basic X8 zero)

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS	Ratio
Z	0	1	-30	-40-M	-35	0	0	0	M	0	-10M	
X4	1	0	2	3	2	1	0	0	0	0	120	40
X5	2	0	4	2	3	0	1	0	0	0	160	80
X6	3	0	3	2	4	0	0	1	0	0	140	70
X8	4	0	0	1	0	0	0	0	-1	1	10	10



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X2 - it has the largest (in absolute value) negative coefficient in row 0 (-40-M).
- Minimum ratio test (RHS ÷ positive entries of the X2-column): R1: 40, R2: 80, R3: 70, R4: 10.
- Leaving basic variable: X8 (row 4, smallest ratio). Pivot number = 1.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Pivot number is already 1 - scaling not needed (normally divide the pivot row by the pivot number)

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS	Ratio
Z	0	1	-30	-40-M	-35	0	0	0	M	0	-10M	
X4	1	0	2	3	2	1	0	0	0	0	120	40
X5	2	0	4	2	3	0	1	0	0	0	160	80
X6	3	0	3	2	4	0	0	1	0	0	140	70
X8	4	0	0	1	0	0	0	0	-1	1	10	10

(b) X2 replaced X8 in the basis. Z = 400

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS
Z	0	1	-30	0	-35	0	0	0	-40	40+M	400
X4	1	0	2	0	2	1	0	0	3	-3	90
X5	2	0	4	0	3	0	1	0	2	-2	140
X6	3	0	3	0	4	0	0	1	2	-2	120
X2	4	0	0	1	0	0	0	0	-1	1	10

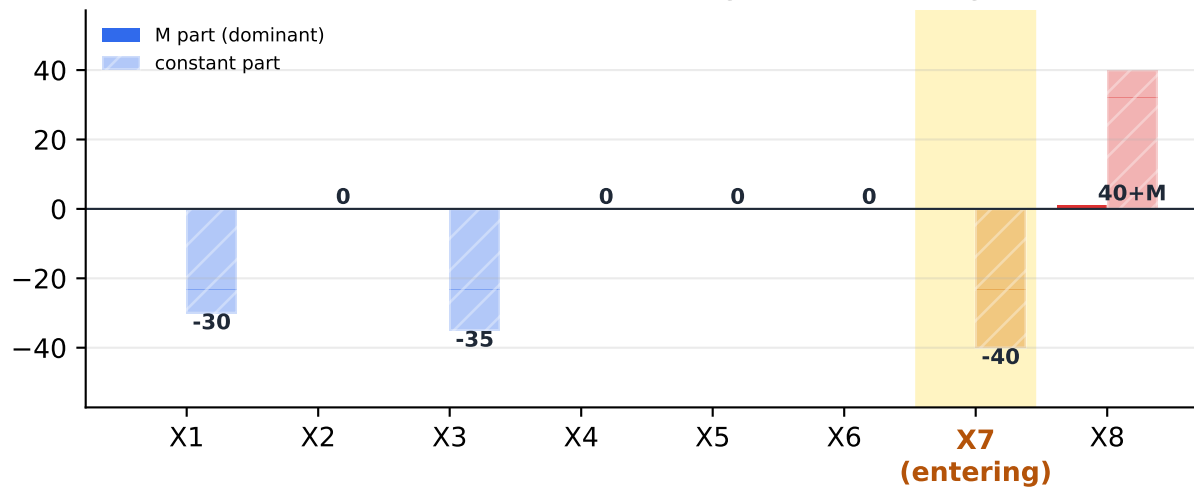
Row operations:

- Add $(40+M) \times R4$ to $R0$
- Subtract $3 \times R4$ from $R1$
- Subtract $2 \times R4$ from $R2$
- Subtract $2 \times R4$ from $R3$

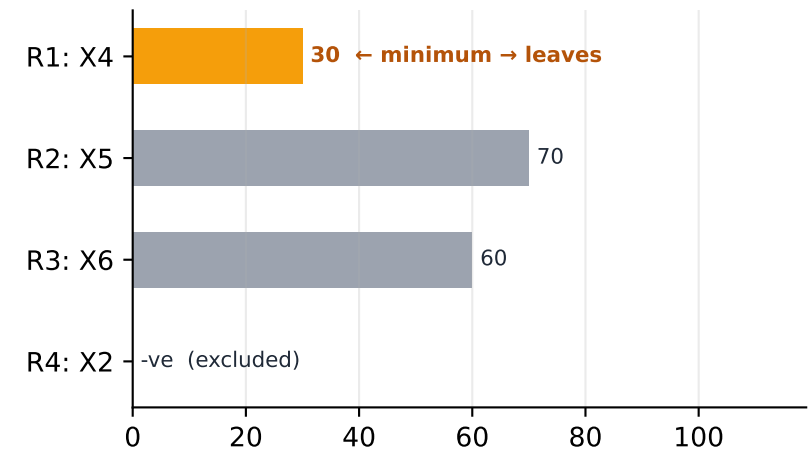
Iteration 2: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS	Ratio
Z	0	1	-30	0	-35	0	0	0	-40	40+M	400	
X4	1	0	2	0	2	1	0	0	3	-3	90	30
X5	2	0	4	0	3	0	1	0	2	-2	140	70
X6	3	0	3	0	4	0	0	1	2	-2	120	60
X2	4	0	0	1	0	0	0	0	-1	1	10	-ve

Row 0 coefficients (most negative → entering)



Minimum ratio test (RHS ÷ X7-column)



Iteration 2 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X7 - it has the largest (in absolute value) negative coefficient in row 0 (-40).
- Minimum ratio test (RHS ÷ positive entries of the X7-column): R1: 30, R2: 70, R3: 60, R4: -ve.
- Leaving basic variable: X4 (row 1, smallest ratio). Pivot number = 3.

Iteration 2: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R1 by the pivot number 3

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS	Ratio
Z	0	1	-30	0	-35	0	0	0	-40	40+M	400	
X4	1	0	0.6667	0	0.6667	0.3333	0	0	1	-1	30	30
X5	2	0	4	0	3	0	1	0	2	-2	140	70
X6	3	0	3	0	4	0	0	1	2	-2	120	60
X2	4	0	0	1	0	0	0	0	-1	1	10	-ve

(b) X7 replaced X4 in the basis. Z = 1600

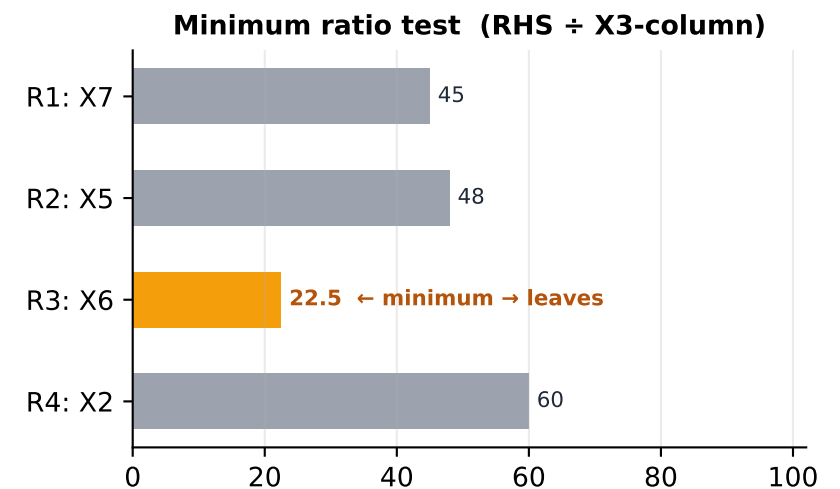
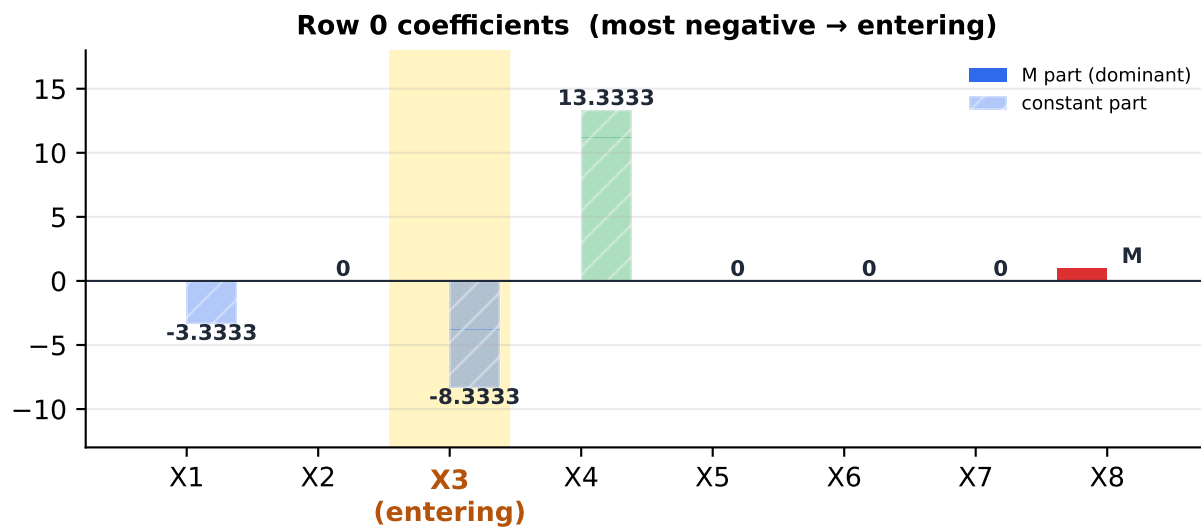
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS
Z	0	1	-3.3333	0	-8.3333	13.3333	0	0	0	M	1600
X7	1	0	0.6667	0	0.6667	0.3333	0	0	1	-1	30
X5	2	0	2.6667	0	1.6667	-0.6667	1	0	0	0	80
X6	3	0	1.6667	0	2.6667	-0.6667	0	1	0	0	60
X2	4	0	0.6667	1	0.6667	0.3333	0	0	0	0	40

Row operations:

- Add $40 \times R1$ to R0
- Subtract $2 \times R1$ from R2
- Subtract $2 \times R1$ from R3
- Add $1 \times R1$ to R4

Iteration 3: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS	Ratio
Z	0	1	-3.3333	0	-8.3333	13.3333	0	0	0	M	1600	
X7	1	0	0.6667	0	0.6667	0.3333	0	0	1	-1	30	45
X5	2	0	2.6667	0	1.6667	-0.6667	1	0	0	0	80	48
X6	3	0	1.6667	0	2.6667	-0.6667	0	1	0	0	60	22.5
X2	4	0	0.6667	1	0.6667	0.3333	0	0	0	0	40	60



Iteration 3 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X3 - it has the largest (in absolute value) negative coefficient in row 0 (-8.3333).
- Minimum ratio test (RHS ÷ positive entries of the X3-column): R1: 45, R2: 48, R3: 22.5, R4: 60.
- Leaving basic variable: X6 (row 3, smallest ratio). Pivot number = 2.6667.

Iteration 3: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R3 by the pivot number 2.6667

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS	Ratio
Z	0	1	-3.3333	0	-8.3333	13.3333	0	0	0	M	1600	
X7	1	0	0.6667	0	0.6667	0.3333	0	0	1	-1	30	45
X5	2	0	2.6667	0	1.6667	-0.6667	1	0	0	0	80	48
X6	3	0	0.625	0	1	-0.25	0	0.375	0	0	22.5	22.5
X2	4	0	0.6667	1	0.6667	0.3333	0	0	0	0	40	60

(b) X3 replaced X6 in the basis. Z = 1787.5

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS
Z	0	1	1.875	0	0	11.25	0	3.125	0	M	1787.5
X7	1	0	0.25	0	0	0.5	0	-0.25	1	-1	15
X5	2	0	1.625	0	0	-0.25	1	-0.625	0	0	42.5
X3	3	0	0.625	0	1	-0.25	0	0.375	0	0	22.5
X2	4	0	0.25	1	0	0.5	0	-0.25	0	0	25

Row operations:

- Add $8.3333 \times R3$ to R0
- Subtract $0.6667 \times R3$ from R1
- Subtract $1.6667 \times R3$ from R2
- Subtract $0.6667 \times R3$ from R4

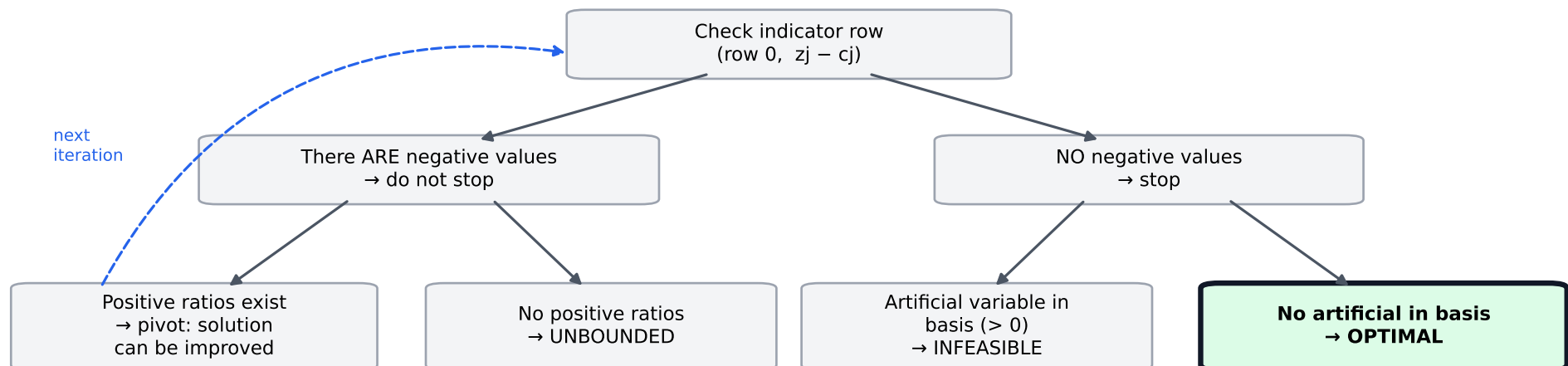
Final tableau - OPTIMAL

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	X7	X8	RHS
Z	0	1	1.875	0	0	11.25	0	3.125	0	M	1787.5
X7	1	0	0.25	0	0	0.5	0	-0.25	1	-1	15
X5	2	0	1.625	0	0	-0.25	1	-0.625	0	0	42.5
X3	3	0	0.625	0	1	-0.25	0	0.375	0	0	22.5
X2	4	0	0.25	1	0	0.5	0	-0.25	0	0	25

Final check.

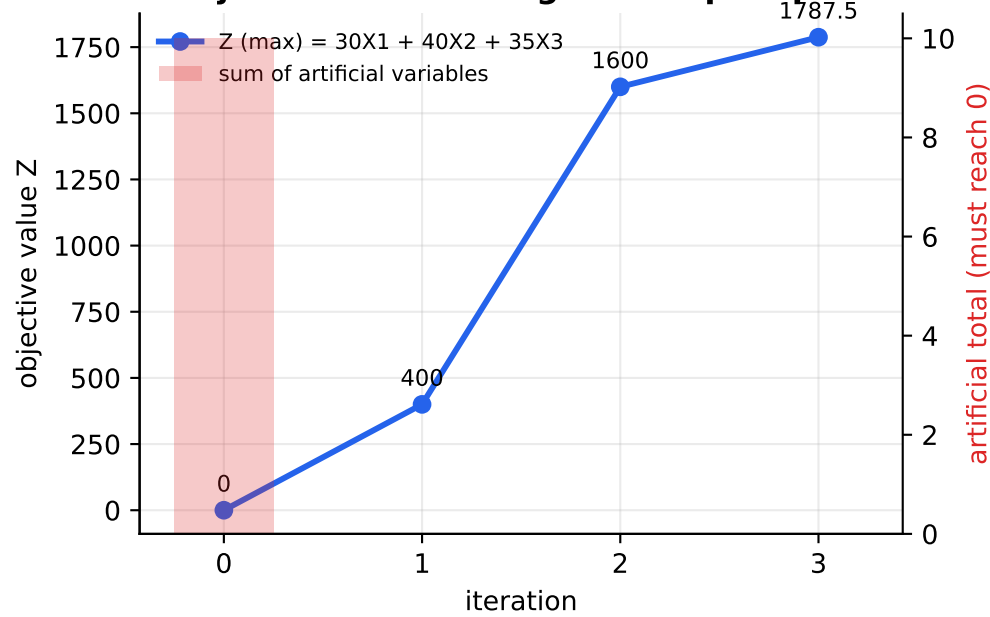
- Optimal: row 0 has no negative coefficients. $Z^* = 1787.5$

Simplex decision diagram - outcome: OPTIMAL

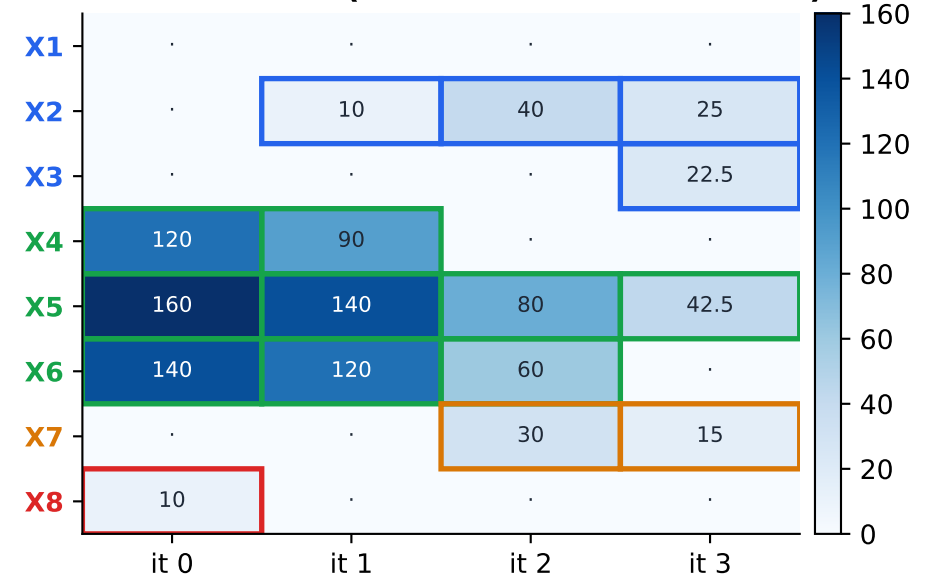


Practice - Three-Product Production Plan - OPTIMAL, $Z^* = 1787.5$

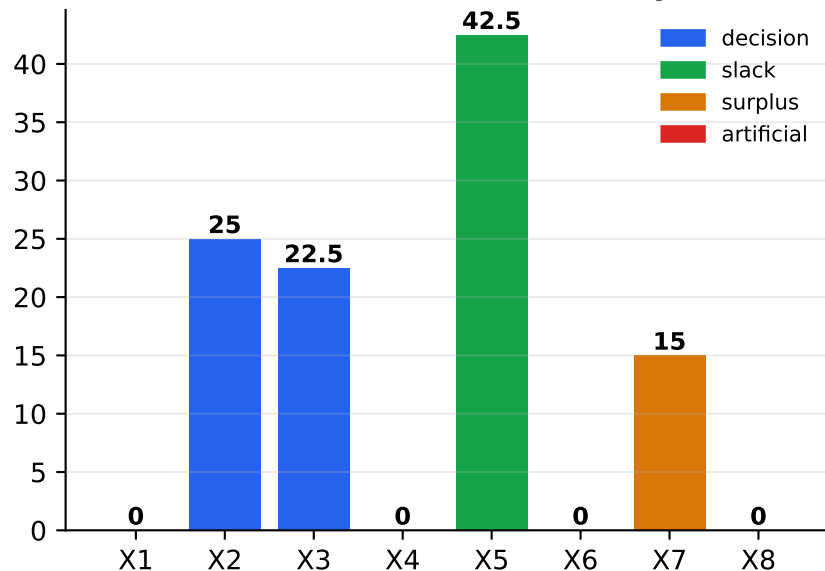
Objective value along the simplex path



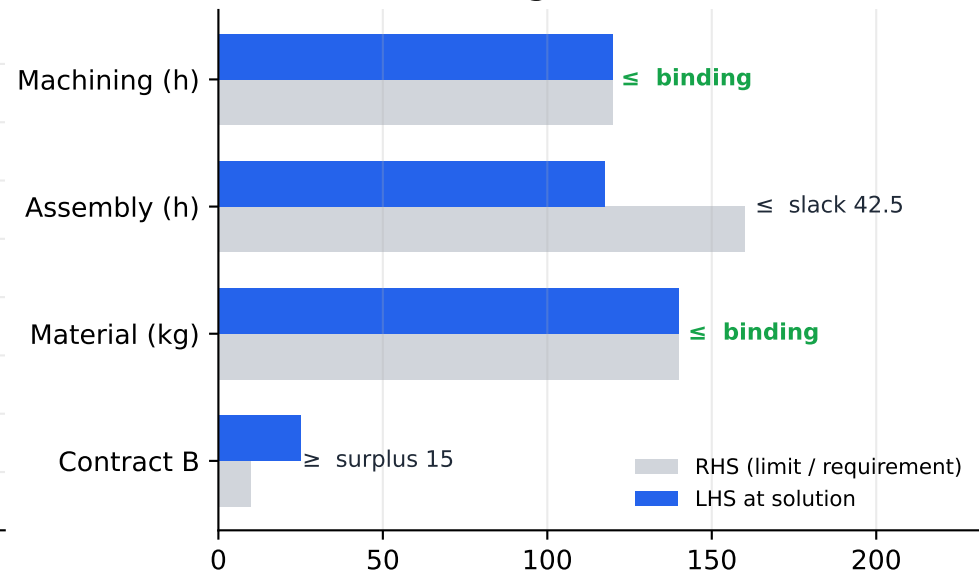
Basis evolution (outlined = basic variable)



Final values of all variables (optimal)



Constraint usage at the final solution



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	optimal	optimal
Z	1787.5	1787.5
X1	0	0
X2	25	25
X3	22.5	22.5

✓ **Results agree**